

Homotopy Invariance

Algebraic Topology I | Fall 2026 | October 5 and 7

Use Lecture 6. Give proofs, state the source and target of every map you define, and keep the signs in chain calculations explicit. Coefficients are integers. Write $I = [0, 1]$ and use all singular simplices, including degenerate ones.

Exercise 1. The Homotopy Category and Homotopy Invariant Functors

A homotopy from $f : X \rightarrow Y$ to $g : X \rightarrow Y$ is a continuous map $h : X \times I \rightarrow Y$ with $h(x, 0) = f(x)$ and $h(x, 1) = g(x)$. A homotopy inverse of f is a map $g : Y \rightarrow X$ with $gf \simeq \text{id}_X$ and $fg \simeq \text{id}_Y$. A nonempty space is contractible if its identity map is homotopic to a constant map.

The category hTop has spaces as objects and homotopy classes of continuous maps as morphisms. The quotient functor is $q : \text{Top} \rightarrow \text{hTop}$. For a category $\mathcal{C}$, a functor $F : \text{Top} \rightarrow \mathcal{C}$ is homotopy invariant if $F(f) = F(g)$ whenever $f \simeq g$.

1. Prove that composition in hTop is well defined. Prove that F is homotopy invariant if and only if there is a unique functor $\bar{F} : \text{hTop} \rightarrow \mathcal{C}$ with $F = \bar{F}q$.
2. Prove that $[f]$ is an isomorphism in hTop if and only if f is a homotopy equivalence. Deduce that every homotopy invariant functor sends homotopy equivalences to isomorphisms, and identify the inverse of $F(f)$.
3. For $\epsilon \in \{0, 1\}$ define $j_\epsilon : X \rightarrow X \times I$ by $j_\epsilon(x) = (x, \epsilon)$, and let $p : X \times I \rightarrow X$ be projection. Show that F is homotopy invariant if and only if $F(p)$ is an isomorphism for every X . In the reverse implication, first show that $F(j_0) = F(j_1)$.

Exercise 2. The Prism over a Singular Edge

The singular group $C_n(X)$ is free abelian on continuous maps $\sigma : |\Delta^n| \rightarrow X$. Write σ also for its generator. Here

$$|\Delta^n| = \{(t_0, \dots, t_n) \in \mathbb{R}^{n+1} : t_i \geq 0, \sum_i t_i = 1\},$$

with standard vertices $e_0, \dots, e_n$. The affine map $d^i : |\Delta^{n-1}| \rightarrow |\Delta^n|$ inserts a zero in position i , and $\partial\sigma = \sum_i (-1)^i \sigma d^i$ for $n > 0$. Let $h : X \times I \rightarrow Y$ be a homotopy from f to g , and let $\sigma : |\Delta^1| \rightarrow X$ be a singular edge. The affine maps $\tau_0, \tau_1 : |\Delta^2| \rightarrow |\Delta^1| \times I$ have ordered vertices

$$\tau_0 : [(e_0, 0), (e_0, 1), (e_1, 1)], \quad \tau_1 : [(e_0, 0), (e_1, 0), (e_1, 1)].$$

Define $P_1 : C_1(X) \rightarrow C_2(Y)$ on generators by

$$P_1(\sigma) = h(\sigma \times \text{id}_I)\tau_0 - h(\sigma \times \text{id}_I)\tau_1.$$

For a point $x \in X$, let $P_0(x) : |\Delta^1| \rightarrow Y$ be the path $P_0(x)(t_0, t_1) = h(x, t_1)$, extended linearly to $P_0 : C_0(X) \rightarrow C_1(Y)$.

1. Draw the square and its diagonal. Write both affine maps in coordinates (t_0, t_1, t_2) , including their two target coordinates in $|\Delta^1| \times I$.
2. List the three ordered edges of each triangle, with their coefficients in $\partial P_1(\sigma)$. Identify the top, bottom, side, and interior edges. Explain geometrically why the diagonal is interior and the two vertical edges are side edges.

3. Verify directly that

$$\partial P_1(\sigma) + P_0(\partial \sigma) = g_{\#}(\sigma) - f_{\#}(\sigma).$$

In particular, exhibit the two equal singular edges on the diagonal and their opposite coefficients.

Exercise 3. Explicit Homotopies and Homology

A nonempty subset $X \subseteq \mathbb{R}^m$ is star shaped about $p \in X$ if $(1-t)x + tp \in X$ for every $x \in X$ and $t \in I$.

1. Construct a contraction of a star shaped space. Compute all its singular homology groups, including degree zero. Explain why the same argument applies to every nonempty convex subset.
2. For $m \geq 1$, define $r : \mathbb{R}^m \setminus \{0\} \rightarrow S^{m-1}$ by $r(x) = x/\|x\|$, and let $i : S^{m-1} \hookrightarrow \mathbb{R}^m \setminus \{0\}$ be inclusion. Construct a homotopy from id to ir that fixes the sphere throughout. Deduce that i_* and r_* are inverse in every degree. No computation of the homology of a positive-dimensional sphere is required.
3. Take $m = 1$ and compute every homology group of $\mathbb{R} \setminus \{0\}$. Explain why the punctured line cannot be homotopy equivalent to the line.

Exercise 4. Retracts and Chain Homotopies

A retraction of a subspace $X' \subseteq X$ is a map $r : X \rightarrow X'$ with $ri = \text{id}_{X'}$, where $i : X' \hookrightarrow X$ is inclusion.

1. Prove that $i_* : H_n(X') \rightarrow H_n(X)$ is injective and r_* is surjective. If X is contractible and X' is nonempty, use a contraction of X to construct a contraction of X' .
2. Let $u, v : A_{\bullet} \rightarrow B_{\bullet}$ be chain maps and let $K_n : A_n \rightarrow B_{n+1}$ satisfy $v - u = d_B K + K d_A$. Show explicitly that u and v induce the same map on homology.
3. Suppose $u : A_{\bullet} \rightarrow B_{\bullet}$ has a chain map $w : B_{\bullet} \rightarrow A_{\bullet}$ such that wu and uw are chain homotopic to the respective identities. Identify the inverse of u_* , and explain how the prism construction applies this algebraic fact to a homotopy equivalence of spaces.

Exercise 5. The Interval and Its Endpoints

For a subspace $X' \subseteq X$, put $C_{\bullet}(X, X') = C_{\bullet}(X)/C_{\bullet}(X')$ and $H_n(X, X') = H_n(C_{\bullet}(X, X'))$.

Let $X = I$ and $X' = \{0, 1\}$. Use the long exact sequence associated to

$$0 \rightarrow C_{\bullet}(X') \rightarrow C_{\bullet}(X) \rightarrow C_{\bullet}(X, X') \rightarrow 0.$$

1. Compute $H_n(X')$ and $H_n(X)$ for all n . Identify the map $H_0(X') \rightarrow H_0(X)$ under the identifications with $\mathbb{Z}^2$ and $\mathbb{Z}$.
2. Deduce all the relative groups $H_n(X, X')$. For the singular edge $\sigma : |\Delta^1| \rightarrow I$ given by $\sigma(t_0, t_1) = t_1$, compute its relative class and its image under the connecting map to $H_0(X')$. Distinguish the two point generators from integer coefficients.
3. The reflection $f : I \rightarrow I$, $f(x) = 1 - x$, preserves X' . Compute its induced map on $H_1(X, X')$. Show that f is homotopic to id_I as a map of spaces, but that there is no such homotopy through maps preserving X' at every time. Explain why there is no contradiction with relative homotopy invariance.

Exercise 6. A Contractible Space with Several Marked Points

Let X be a contractible space and let $X' = \{x_0, \dots, x_{k-1}\} \subseteq X$ be a discrete subspace of $k \geq 1$ distinct points.

1. Compute all the groups $H_n(X, X')$. Identify $H_1(X, X')$ canonically with the kernel of the homomorphism $\mathbb{Z}^k \rightarrow \mathbb{Z}$ that adds coordinates.
2. Choose paths $\sigma_i : |\Delta^1| \rightarrow X$ from x_0 to x_i for $1 \leq i < k$. Prove that their relative homology classes form a basis of $H_1(X, X')$. Prove that these classes are independent of the chosen paths with the same endpoints.
3. Suppose $f : X \rightarrow X$ maps X' to itself, with $f(x_i) = x_{\pi(i)}$ for a permutation π . Describe the induced action on the kernel above and give its matrix in the basis from part 2. Recover the sign of reflection in the preceding exercise when $k = 2$.